

ASIM ALTAF

Backend Engineer — Go · C/C++ · Distributed & Telecom Systems

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Portfolio: asim-altaf-portfolio.vercel.app · GitHub: github.com/AsimAltaf-eg

PROFILE

Backend engineer specializing in **high-performance, production-grade systems** built in **Go** and **C/C++** — spanning telecom signalling, subscriber data, and distributed infrastructure. At Safarifone, I work on the authentication, eSIM provisioning, and **5G core** systems supporting a **multi-million-subscriber network**. Previously, I built VoIP/SIP infrastructure handling **10,000+ calls per second**. I care about performance, correctness, and systems that hold up reliably under real-world load.

EXPERIENCE

Backend / Systems Engineer · Safarifone (mobile network operator)

Nov 2025 – Present · Islamabad

Work across four production teams at multi-million subscriber scale: subscriber authentication (5G core), eSIM provisioning, SS7 signalling, and the subscriber platform.

- Own subscriber authentication for our 3GPP UDM/UDR (the 5G-generation successor to the HSS): implemented **SUPI/SUCI resolution and authentication-vector generation** to TS 29.503/29.505/23.502 in C with libuv — conceptually the same problem as issuing and validating credentials at scale, applied within the telecom domain.
- Replaced a thread-heavy design in that authentication path with an **async event loop**, reducing **p95/p99 latency measurably**. Profiled a **50K+ line C codebase** with Valgrind to ship the rewrite without regressions.
- Architected our eSIM provisioning service in Go against a live SM-DP+ (GSMA SGP.02, ES2+): built the **mTLS client-authentication layer**, callback handling, and the full profile-issuance state machine end to end, from order through installed profile.
- Built the HLR/MAP adapter (SendRoutingInfo, UpdateLocation, InsertSubscriberData) and a TCAP session/transaction manager for USSD over SIGTRAN, in Go — including a **custom zero-allocation binary framing protocol** and the SS7-to-JSON translation layer that other engineering teams rely on.
- Day to day: **Go**, **C/C++**, **gRPC/Protobuf**, **Kubernetes**, **Docker**, **Temporal.io**, **PostgreSQL**, **Oracle**, **Redis**.

Golang Developer (part-time remote contract) · ContactVA — US enterprise clients

Sep 2024 – Jul 2026 · Remote

- Built VoIP/SIP infrastructure in Go: a UDP signalling server sustaining **10,000+ calls per second**, with SIP signalling, RTP relay, and OpenSIPS call routing — hands-on experience operating network-facing Go services under continuous, real-world load.
- **Reduced API latency from ~1,000ms to ~300ms** by reworking OpenSIPS flows and tuning the PostgreSQL/Redis data layer.
- Ran alongside the Safarifone role from late 2025, with the client's agreement, until the contract concluded in July 2026.

Backend Developer · ZIXEL

Sep 2023 – Jun 2024 · Islamabad

- First role out of university, building and maintaining **Go microservices across five products**.
- Introduced caching and autoscaling improvements that **reduced response times by roughly 20%**; mentored two junior backend developers.

TECHNICAL SCOPE

- **Core systems:** **Go** (production, multi-year), **C**, **C++** (libuv), concurrency & event-loop design, **gRPC/Protobuf**.
- **Signalling & telecom:** **SS7** (MAP, TCAP, USSD, SIGTRAN), **HLR/HSS**, **5G SBA** (UDM/UDR), **SIP/RTP**, **eSIM** (GSMA SGP.02, ES2+).
- **Security & authentication:** cryptographic authentication-vector generation (SUPI/SUCI), **mTLS client authentication**, session/transaction state management (TCAP).
- **Data & infra:** **PostgreSQL**, **Oracle**, **Redis**, **Kubernetes**, **Docker**, **Temporal.io**.

EDUCATION

BS Computer Science — National University of Computer & Emerging Sciences (FAST-NUCES)

2019 – 2023

- Final Year Project selected for a funded incubation program, supporting further development and commercialization.

CERTIFICATIONS

5G Network Fundamentals · Go — Complete Course · AWS Certified Cloud Practitioner (in progress)